

Shikhar Misra

LTD Yield Engineer
Defect Metrology Group
Intel Corporation

RA4-601
Ronler Acres Campus 4
Hillsboro, OR 97124
(765)-775-6573

✉ shikhar.misra@intel.com, shikharmisra94@gmail.com
🌐 <https://www.linkedin.com/in/shikhar-misra-a69aa640/>

Education

- 2016-2020 **Purdue University, West Lafayette, Indiana, USA.**
Doctor of Philosophy (PhD) in *Materials Engineering* GPA: **3.87/4**
Advisor: Dr. Haiyan Wang
Research Interests: Thin Film Epitaxy, Nanocomposites, Metamaterials
- 2015-2016 **Indian Institute of Technology Kanpur, India.**
Masters of Technology (M. Tech) in *Materials Science and Engineering* GPA: **10/10**
Advisor: Dr. Krishanu Biswas
- 2011-2015 **Indian Institute of Technology Kanpur, India.**
Bachelors of Technology (B. Tech) in *Materials Science and Engineering* GPA: **8.2/10**

Honors and Awards

- 2019 Won the **Best Poster** Award (top 1%) at the Materials Research Society (**MRS**) Fall Meeting and Exhibit, 2019
- 2019 Awarded the PGSG travel grant (top 10%) and the College of Engineering **travel grant**
- 2019-20 Recipient of the highly competitive **Bilsland Dissertation Fellowship** provided by the Graduate School given to one student from the MSE department at Purdue University
- 2016 Recipient of prestigious **IIT-Todai** graduate student scholarship 2016 among 10 students from India
- 2014, 2015 **Academic Excellence Award** recipient for the consecutive academic year 2013-14 and 2014-2015, awarded to top 5% students at IIT Kanpur
- 2011 Secured **All India Rank 2300** (Percentile:99.54) in IIT-JEE'11 and **All India Rank 937** in AIEEE'11
- 2011-16 Awarded **scholarship** 'Incentive scheme for meritorious children' by State Bank of India for being in good academic standing throughout the stay at IIT Kanpur
- 2010 **7th position** in Regional Mathematics Olympiad (RMO 2010) in the state of Uttar Pradesh

Professional Experience

- Oct'20– present **Intel Corporation, Hillsboro, Oregon, USA.**
Title : Technology Development (TD) Module and Integration Yield Engineer (Defect Metrology)
- Logic Technology Development (LTD) yield engineer in the defect metrology module developing 10, 7, and 5 nm process technologies
 - Drove the development of the process for the unpatterned wafer inspection scan and review at the front end of the manufacturing process
 - Close collaboration with cross functional teams of Process and Integration engineers to determine technology requirements
 - Transfer of the technology to Intel's 300 mm factories in Arizona, Israel and Ireland
 - Partnered with equipment suppliers to drive the development of the hardware and process
 - Routinely applied SPC, JMP and statistical Data Analytics to monitor trends and find causes for signals to improve device yield

Research Experience

Aug'16– Aug'20 **Metal-oxide nanocomposites for tunable physical properties**, *Graduate Research Assistant, Purdue University.*

Advisor : Prof. Haiyan Wang

Objective: Exploration of multiple metal-oxide thin film nanocomposite systems to study the light-matter interaction. The goal span both tuning the physical properties and understanding the growth mechanisms of complex nanocomposite systems towards the successful fabrication of nanocomposite-based devices.

Aug'15– May'16 **Bulk Preparation of Graphene: Synthesis and Applications**, *M.Tech Thesis, IIT Kanpur, India.*

Advisor : Prof. Krishanu Biswas

Objective: Fabricating graphene and graphene-silver nanoparticle hybrid cheaply on a large scale using a domestic kitchen blender for water purification applications

Summer'15 **Current injection in heterostructures based on 2D semiconductors**, *EPFL, Switzerland.*

Advisor : Prof. Andras Kis

Objective: To demonstrate spin injection from a ferromagnetic electrode into a heterojunction based on monolayers of WSe_2 and MoS_2

Summer'14 **Growth of CVD graphene with mixed ^{12}C and ^{13}C isotopes**, *University of Vienna, Austria.*

Advisor : Prof. Jannik Meyer

Objective: To decipher the CVD growth mechanism of benzene by synthesizing mixed graphene followed by analysis in STEM using EELS spectroscopy

Summer'13 **Refinement of austenite grain structure of Fe-0.45C Steel**, *Hokkaido University, Japan.*

Advisor : Prof. Munekazu Ohno

Objective: To refine inversely transformed austenite grain structure of Fe-0.45C Steel by Nb addition

Technical Skills

Fabrication	Pulsed Laser Deposition (PLD), Chemical Vapor Deposition (CVD), Mechanical exfoliation
Characterization	Atomic Force Microscopy (AFM), Piezoresponse Force Microscopy (PFM), Transmission Electron Microscopy (TEM), X-Ray Diffraction (XRD), Raman Spectroscopy, UV-Vis Spectroscopy, Ellipsometry, Physical Property Measurement System (PPMS)
Computation	COMSOL Multiphysics, Tensorflow, Keras, ANSYS Fluent, JMP, AutoCAD, DesignCAD
Graphics & Design	Adobe Illustrator, Adobe Premiere Pro, Adobe After Effects, Adobe Photoshop, 3DS Max, Inkscape

Journal Publications

32 publications in peer-reviewed journals, including 2 invited reviews, >340 citations, h-index=10

★ First Author

- 2020 1. **S. Misra**, D. Zhang, P. Lu, H. Wang *Thermal stability of self-assembled ordered three-phase $Au - BaTiO_3 - ZnO$ nanocomposite thin films via in situ heating in TEM*, published in [Nanoscale](#), (2020)
2. **S. Misra**, D. Zhang, Z. Qi, D. Li, J. Lu, H.-T. Chen, H. Wang *Morphology Control of Self-Assembled Three-Phase $Au - BaTiO_3 - ZnO$ Hybrid Metamaterial for Tunable Optical Properties*, published in [Crystal Growth and Design](#), 20, 6101-6108 (2020)
3. **S. Misra**, M. Kalaswad, D. Zhang, H. Wang, *Dynamic measurement of electrically tunable dielectric permittivity in $BaTiO_3$ using spectroscopic ellipsometer*, published in [Materials Research Letters](#), 8, 321-327 (2020)

4. **S. Misra**, L. Li, X. Gao, J. Jian, Z. Qi, D. Zemlyanov, H. Wang L. Li, D. Zhang, J. Jian, Z. Qi, M. Fan, H.-T. Chen, X. Zhang, H. Wang, *Tunable Physical Properties in $Bi_2Al_{1-x}Mn_xO_6$ Thin Films with Novel Layered Supercell Structures*, published in [Nanoscale Advances](#), 2, 315-322 (2020)
- 2019 5. **S. Misra**, L. Li, D. Zhang, J. Jian, Z. Qi, M. Fan, H.-T. Chen, X. Zhang, H. Wang, *Self-assembled ordered three-phase $Au - BaTiO_3 - ZnO$ vertically aligned nanocomposites achieved by a templating method*, published in [Advanced Materials](#), 31(7), 1806529 (2019)
- 2018 6. **S. Misra**, L. Li, J. Jian, J. Huang, X. Wang, D. Zemlyanov, J. Jang, F.H. Ribeiro, H. Wang, *Tailorable Au nanoparticles embedded in epitaxial TiO_2 thin films for efficient photocatalysis*, published in [ACS Applied Materials and Interfaces](#), 10(6), 5779-5784 (2018)

★ Invited Review Articles

- 2021 7. O.J. Lee*, **S. Misra***, H. Wang, J.L. Mac-Manus Driscoll, *Ferroelectric/multiferroic self-assembled vertically aligned nanocomposites: Current and future status*, published in [APL Materials](#), (2021)
8. **S. Misra**, H. Wang, *Review on the growth, properties and applications of self-assembled oxide-metal vertically aligned nanocomposite thin films—current and future perspectives*, published in [Materials Horizon](#), (2021)

★ Co-Author

- 2021 9. R.L. Paldi, Z. Qi, **S. Misra**, J. Lu, X. Sun, X.L. Phuah, M. Kalaswad, J. Bischoff, D.W. Branch, A. Siddiqui, H. Wang *Nanocomposite-Seeded Epitaxial Growth of Single-Domain Lithium Niobate Thin Films for Surface Acoustic Wave Devices*, published in [ACS Applied Materials and Interfaces](#), (2021)
10. Z. Qi, R. Xu, **S. Misra**, H. Wang, J. Huang, K. Zhao, *Ellipsometry-based failure analysis on translucent $LiMn_{0.5}Ni_{0.3}Co_{0.2}O_2$ in half-cell thin-film lithium-ion battery on glass substrates*, published in [ACS Applied Materials and Interfaces](#), (2021)
11. D. Zhang, **S. Misra**, J. Jian, P. Lu, L. Li, A. Wissel, X. Zhang, H. Wang *Self-Assembled $BaTiO_3/Au_xAg_{1-x}$ Low-Loss Hybrid Plasmonic Metamaterials with an Ordered “Nano-Domino-like” Microstructure*, published in [ACS Applied Materials and Interfaces](#), (2021)
- 2020 12. D. Zhang, P. Lu, **S. Misra**, A. Wissel, Z. He, Z. Qi, X. Gao, X. Sun, J. Liu, J. Lu, X. Zhang, H. Wang, *Design of 3D Oxide-Metal Hybrid Metamaterial for Tailorable Light-Matter Interactions in Visible and Near-Infrared Region*, published in [Advanced Optical Materials](#), (2020)
13. Z. He, J. Jian, **S. Misra**, X. Gao, X. Wang, Z. Qi, D. Zhang, H. Wang, *Bidirectional tuning of phase transition properties in $Pt : VO_2$ nanocomposite thin films*, published in [Nanoscale](#), 12, 17886-17894 (2020)
14. J. Liu, X. Wang, X. Gao, H. Wang, J. Jian, J. Huang, X. Sun, Z. Qi, **S. Misra**, Z. He, H. Wang, *Multifunctional self-assembled $BaTiO_3 - Au$ nanocomposite thin films on flexible mica substrates with tunable optical properties*, published in [Applied Materials Today](#), 21, 100856 (2020)
15. B. Zhang, M. Kalaswad, B. Rutherford, **S. Misra**, Z. He, H. Wang, Z. Qi, A. Wissel, X. Xu, H. Wang, *Au-Encapsulated Fe Nanorods in Oxide Matrix with Tunable Magneto-Optic Coupling Properties*, published in [ACS Applied Materials and Interfaces](#), 12, 51827-51836 (2020)
16. B. Zhang, J. Huang, B. Rutherford, P. Lu, **S. Misra**, M. Kalaswad, X. Sun, L. Li, H. Wang, *Tunable, Room Temperature Multiferroic $Fe - BaTiO_3$ Vertically Aligned Nanocomposite with Perpendicular Magnetic Anisotropy*, published in [Materials Today Nano](#), 100083 (2020)
17. Z. Qi, J. Tang, **S. Misra**, C. Fan, P. Lu, J. Jian, Z. He, V. G. Pol, X. Zhang, H. Wang, *Enhancing electrochemical performance of thin film lithium ion battery via introducing tilted metal nanopillars as effective current collectors*, published in [Nano Energy](#), 69, 104381 (2020)
18. X.L. Phuah, H. Wang, Z. Qi, **S. Misra**, M. Kalaswad, H. Wang, *Flash sintering of Gd-doped ceria thin film*, published in [Journal of the American Ceramics Society](#), 103, 2309-2314 (2020)

*equal contribution

- 2019 19. Y. Ji, Z. Qi, **S. Misra**, R. Jin, Y. Liu, X. Ou, Y. Lin, H. Yang, H. Wang, *Breaking lattice symmetry in highly strained epitaxial VO₂ films on faceted nanosurface*, published in [ACS Applied Materials and Interfaces](#), 11, 44905-44912 (2019)
20. D. Zhang, **S. Misra**, L. Li, X. Wang, X. Gao, X. Sun, J. Jian, P. Lu, Z. Qi, M. Kalaswad, X. Zhang, H. Wang, *Optical Tuning of Self-Assembled Au-BaTiO₃ Vertically Aligned Nanocomposites via Geometry Control: from Nanopillar to Nanodisk*, published in [Advanced Optical Materials](#), 1901359 (2019)
21. L. Li, J. Cheng, H. Wang, J. Huang, X. Gao, X. Wang, **S. Misra**, B. Zhang, J. Jian, A. Chen, P. Lu, X. Qian, K. Yang, H. Wang, *Interfacial Engineering Enabled Novel Bi-Based Layered Oxide Supercells with Modulated Microstructures and Tunable Physical Properties*, published in [Crystal Growth and Design](#), 6, 100037 (2019)
22. B. Zhang, J. Huang, J. Jian, B. Rutherford, L. Li, **S. Misra**, X. Sun, H. Wang, *Tuning Magnetic Anisotropy in Co-BaZrO₃ Vertical Aligned Nanocomposites for Memory Device Integration*, published in [Nanoscale Advances](#), 1 (11), 4450-4458 (2019)
23. J. Jian, X. Wang, **S. Misra**, X. Sun, Z. Qi, X. Gao, J. Sun, A. Donohue, D. G. Lin, V. Pol, J. Youngblood, H. Wang, L. Li, J. Huang and H. Wang, *Broad range tuning of phase transition property in VO₂ through metal-ceramic nanocomposite design*, published in [Advanced Functional Materials](#), 1903690 (2019)
24. L. Li, P. Boullay, J. Cheng, P. Lu, X. Wang, G. Steciuk, J. Huang, J. Jian, X. Gao, B. Zhang, **S. Misra**, X. Zhang, K. Yang, H. Wang, *Self-assembled two-dimensional layered oxide supercells with modulated layer stacking and tunable physical properties for flexible nanoscale electronics*, published in [Materials Today Nano](#), 6, 100037 (2019)
25. J. Wu, B. Gautam, M.A. Sebastian, V. Ogunjimi, **S. Misra**, J. Huang, J. Baca, J. Prestigiacomo, T. Haugan, H. Wang, M. Osofsky, *Pinning Efficiency of one-dimensional artificial pinning centers in YBa₂Cu₃O_{7-x} thin films*, published in [IEEE Transactions on Applied Superconductivity](#), 29(5), 1-5 (2019)
26. M. Parekh, V. Parikh, P. Kim, **S. Misra**, Z. Qi, H. Wang, V. Pol, *Encapsulation and networking of silicon nanoparticles using amorphous carbon and graphite for high performance Li-ion batteries*, published in [Carbon](#), 148, 36-43 (2019)
27. R. Xu, Y. Ji, R. Bouchilaoun, F. Qian, M. Li, X. Zhang, R. Tang, R. Zhao, **S. Misra**, H. Wang, W. Li, C. Kan, D. Shi, J. Fan, H. Yang, *Optical and electrical properties of (111)-oriented epitaxial SrVO₃ thin films*, published in [Ceramics International](#), 45(9), 11304-11308 (2019)
- 2018 28. B. Gautam, M.A. Sebastian, S. Chen, **S. Misra**, J. Huang, F.J. Baca, R. Emergo, T. Haugan, Z. Xing, H. Wang, J.Z. Wu, *Probing the effect of interface on vortex pinning efficiency of one-dimensional BaZrO₃ and BaHfO₃ artificial pinning centers in YBa₂Cu₃O_{7-x} thin films*, published in [Applied Physics Letters](#), 113(21), 212602 (2018)
29. J. Huang, T. Jin, **S. Misra**, H. Wang, Z. Qi, Y. Dai, X. Sun, L. Li, J. Okkema, H.-T. Chen, P.-T. Lin, X. Zhang, H. Wang, *Tailorable optical response of Au-LiNbO₃ hybrid plasmonic metamaterial thin films for optical waveguide applications*, published in [Advanced Optical Materials](#), 6(19), 1800510 (2018)
30. M. Fan, H. Wang, **S. Misra**, B. Zhang, Z. Qi, X. Sun, J. Huang, H. Wang, *Microstructure, magnetic and magnetoresistance properties of La_{0.7}Sr_{0.3}MnO₃:CuO nanocomposite thin films*, published in [ACS Applied Materials and Interfaces](#), 10(6), 5779-5784 (2018)
- 2017 31. L. Li, P. Boullay, P. Lu, X. Wang, J. Jian, J. Huang, X. Gao, **S. Misra**, W. Zhang, O. Perez, G. Steciuk, A. Chen, X. Zhang, H. Wang, *Novel layered supercell structure from Bi₂AIMnO₆ for multifunctionalities*, published in [Nano Letters](#), 17(11), 6575-6582 (2017)
- 2016 32. O.L. Sanchez, D. Ovchinnikov, **S. Misra**, A. Allain, A. Kis, *Valley polarization by spin injection in a light-emitting van der Waals heterojunction*, published in [Nano Letters](#), 16(9), 5792-5797 (2016)

Google Scholar Profile [Shikhar Misra](#)

Selected Conferences and Presentations

- 2020 6. **S. Misra**, L. Li, J. Jian, J. Huang, X. Wang, D. Zemlyanov, J. Jang, F. H. Ribeiro and H. Wang, *Self-assembled ordered three-phase Au-BaTiO₃-ZnO vertically aligned nanocomposites achieved by a templating method*, oral presentation at *Electronic Materials and Applications (2020)*, Orlando, FL, USA
- 2019 5. **S. Misra**, L. Li, J. Jian, J. Huang, X. Wang, D. Zemlyanov, J. Jang, F. H. Ribeiro and H. Wang, *Self-assembled ordered three-phase Au-BaTiO₃-ZnO vertically aligned nanocomposites achieved by a templating method*, won **best poster** at *MRS Fall Meeting and Exhibit (2019)*, Boston, MA, USA
4. **S. Misra**, L. Li, J. Jian, J. Huang, X. Wang, D. Zemlyanov, J. Jang, F. H. Ribeiro and H. Wang, *Tailorable Au nanoparticles embedded in epitaxial TiO₂ thin films for efficient photocatalysis*, oral presentation at *Electronic Materials and Applications (2019)*, Orlando, FL, USA
- 2018 3. J. Wu, B. Gautam, M. A. Sebastian, **S. Misra**, J. Huang, S. Chen, J. Prestigiacomo, T. Haugan, H. Wang, M. Osofsky and Z. Xing, *Probing the effect of interface on pinning efficiency of 1D BaZrO₃ and BaHfO₃ artificial pinning centers in YBa₂Cu₃O_{7-x} thin films*, at *Applied Superconductivity Conference (ASC)(2018)*, Seattle, WA, USA
- 2016 2. **S. Misra**, and K. Biswas, *Bulk preparation of graphene: Synthesis and Applications*, poster presentation at *International Conference on Metals and Materials Research (ICMR)(2016)* IISc Bangalore, Bangalore, KA, India .
- 2014 1. **S. Misra**, K. Mishra and N.P. Gurao, *Cryo-cross Rolling of SS304*, secured 3rd position in poster presentation at *Metallix (2014)*, Jadavpur University, Kolkata, WB, India.

Teaching Experience

- Aug–Dec'18 **Teaching Assistant, MSE370**, *Electrical, Optical and Magnetic Properties of Materials*, Purdue University.
- Jan–Apr'16 **Teaching Assistant, MSE314**, *Process Engineering Laboratory*, IIT Kanpur.
- Jan–Apr'16 **NPTEL Teaching Assistant**, *Online course on Phase Diagrams in Materials Science and Engineering*.
- Jul–Dec'15 **Teaching Assistant, MSE626A**, *Transport Phenomena*, IIT Kanpur.
- Jan–Apr'14 **Academic mentor**, *Hall of Residence-3*, IIT Kanpur.

Professional Affiliations

- Material Advantage, Student member
- Materials Research Society (MRS), Student member
- Indian Institute of Technology Kanpur (IITK) Alumni Association

References

Dr. Haiyan Wang
Professor
School of Materials Engineering
Purdue University
West Lafayette, USA
hwang00@purdue.edu

Dr. Eric Kvam
Professor
School of Materials Engineering
Purdue University
West Lafayette, USA
kvam@purdue.edu

Dr. Krishanu Biswas
Professor
Department of Materials Science and Engineering
Indian Institute of Technology Kanpur
Kanpur, India
kbiswas@iitk.ac.in